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Spartan-6 USB-FPGA module 1.11b

and paid licensed IPs to be used

6. Type of processor interfaces used (soft processor or external processor interfaces)

7. Type of memory interfaces used

8. A section about the timing diagram of the major peripheral interfaces such as the processor interface and flash interface

9. Type of FPGA configurations to be used

10. Reset and clock interface planned

11. A brief summary of the estimated resources required for implementation of the logic and I/O pins to be used

12. HDL (VHDL, Verilog, 'C' or mixed) used for RTL coding, tools and version to be used for synthesis, implementation and simulation

To calculate the approximate resources required, go through the IP datasheets for the resources used for each IP, and also calculate the resources used by custom RTL. There is no rule of thumb for calculating resources at this level. These can be calculated approximately based on experience, reviews or analysis. The most important thing is to get the resource requirement reviewed by the hardware team, software team and a third party several times before submitting it to the client.

Step 2: Detailed design document preparation

Once the SRS is approved by the client, the next phase is to make the detailed

design document.

This document should consist of:

1. Brief introduction to the project
2. FPGA part details with proper specifications
3. Detailed block diagram depicting the internal modules of the FPGA design
4. Top-level module block diagram showing input and output ports with their active levels and voltage levels which are connected to the external peripherals, connectors and debug points
5. Hierarchical tree of the modules
6. Each module should have:
 - (i) Detailed explanation of the functionality
 - (ii) Register information
 - (iii) List of input and output ports with source and destination module name, and active level of the signal
 - (iv) A block diagram/digital circuit diagram of finite-state machines indicating how the RTL will be implemented
 - (v) Clock frequency to be used, if a synchronous module is used
 - (vi) Reset logic implementation
 - (vii) File name which will be implemented
 - (viii) Approximate FPGA resource utilisation
 - (ix) Testbench for testing each module independently
7. Input system clock frequency and reset level
8. Explanation of how the internal clock frequencies are derived - us-